

Physical Chemistry (II)

Lecture 09

1차 중간고사 안내 및 복습



1차 중간고사 안내

- 시간: 10월 11일(수) 오후 1:30~오후 2:30
- 장소: 화학관 B101호(홀수 학번), 114호(짝수 학번)
- 계산기 불필요

시험 범위

강의 자료 기준 3강~8강

(1강, 2강은 범위에서 제외)

시험 문제

총 여섯 문제

- 과제 문제 혹은 그 변형(세 문제)
 - 유도라면 중간 단계를 주지 않을 수도 있음
- 수업 시간에 배운 식 유도 및 응용(세 문제)
 - 강의 자료 기준

Overview

Foundations of Quantum Mechanics

(1) wavefunction, (2) measurement, (3) dynamics



Particle
in a box



Simple
harmonic
oscillator



Rigid rotor

Postulates of Quantum Mechanics

1. Wavefunction and its interpretation

- Wavefunction and probability
- Normalization

2. Quantum measurement

- Observables and operators
- Measurement and collapse of wave function
- Uncertainty principle

3. Time evolution of wavefunction: Schrödinger equation



Observables and Operators

- Every **CM observable** has a corresponding **QM operator**

- Position operators: $\hat{x} = x$, $\hat{y} = y$, $\hat{z} = z$

- Momentum operators:

$$\hat{p}_x = \frac{\hbar}{i} \frac{\partial}{\partial x}, \quad \hat{p}_y = \frac{\hbar}{i} \frac{\partial}{\partial y}, \quad \hat{p}_z = \frac{\hbar}{i} \frac{\partial}{\partial z}$$

$\hbar/2\pi$
↓
○

- System energy operator (Hamiltonian):

$$\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + \mathcal{V}(\hat{x}, \hat{y}, \hat{z}, t)$$

Properties of QM Operators

- Eigenvalues are **real**
- Eigenfunctions form a **complete** set

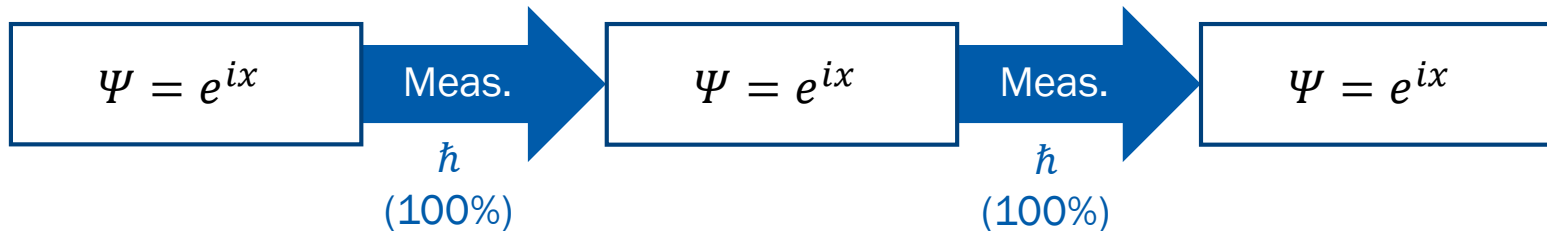
“Any arbitrary function can be represented as a linear combination of the functions from a complete set”
- Eigenfunctions are **orthogonal** to each other

“Two functions are orthogonal to each other if their inner product is zero”

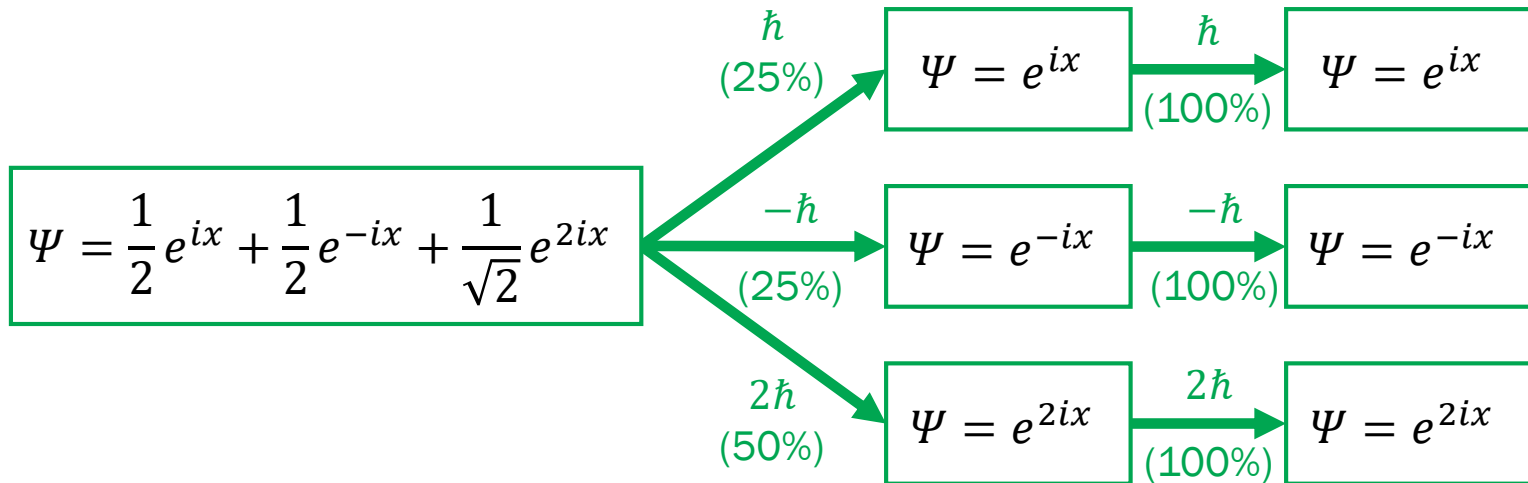
$$\int f^* g \, d\tau = 0$$

Quantum Measurement

- If the wavefunction is the eigenfunction of the operator



- Otherwise



Expectation Value

$$\langle A \rangle = \int \psi^* \hat{A} \psi \, d\tau$$

(Problem Set 1, Problem 3)

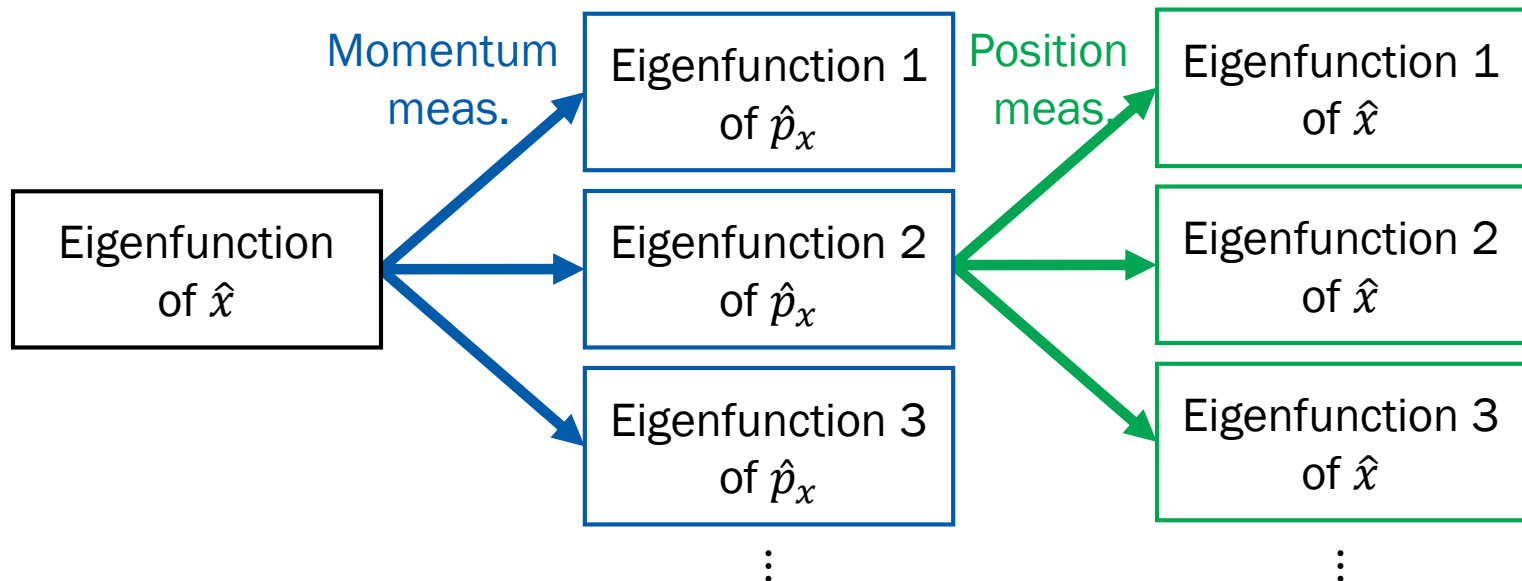
$$\langle A^2 \rangle = \int \psi^* \hat{A} \hat{A} \psi \, d\tau$$

$$\text{Variance: } \sigma_A^2 = \langle A^2 \rangle - \langle A \rangle^2$$

Measurement of Two Observables

The eigenfunction of one QM operator may not be the eigenfunction of another QM operator

(e.g. eigenfunction of $\hat{x} \neq$ eigenfunction of $\hat{p}_x$)



Commutator

- If $\hat{A}\hat{B} = \hat{B}\hat{A}$, $\hat{A}$ and $\hat{B}$ are said to commute
- Commutator

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

- If $\hat{A}$ and $\hat{B}$ commute,

$$[\hat{A}, \hat{B}] = \hat{0}$$

Generalized Uncertainty Principle

$$\sigma_A^2 \sigma_B^2 \geq -\frac{1}{4} \langle [\hat{A}, \hat{B}] \rangle^2$$

e.g. position-momentum uncertainty

$$[\hat{x}, \hat{p}_x] = \hat{x}\hat{p}_x - \hat{p}_x\hat{x} = x \frac{\hbar}{i} \frac{\partial}{\partial x} - \frac{\hbar}{i} \frac{\partial}{\partial x} x = i\hbar$$

$$\sigma_x^2 \sigma_{p_x}^2 \geq -\frac{1}{4} \langle i\hbar \rangle^2 = \left(\frac{\hbar}{2} \right)^2$$

Time Evolution of Wavefunction

$$i\hbar \frac{\partial}{\partial t} \Psi(x, y, z, t) = \hat{H} \Psi(x, y, z, t)$$

where

$$\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + \mathcal{V}(\hat{x}, \hat{y}, \hat{z}, t)$$

(Time-dependent Schrödinger equation)

Time-Independent Solution

- Time-independent Schrödinger equation

$$\hat{H}\psi(x, y, z) = E\psi(x, y, z)$$

→ Eigenvalue equation for the Hamiltonian

- Temporal solution: $T(t) = e^{-iEt/\hbar}$

$$|T(t)|^2 = 1$$

$$|\Psi(x, y, z, t)|^2 = |\psi(x, y, z)|^2$$

(Exactly Solvable) Models

Particle in a box (PIB): 1D, 2D, 3D

Particle on a ring (POR)

Simple harmonic oscillator (SHO)

Rigid rotor (RR)

Four Models

	1D PIB	POR	SHO	RR
Model	L4, 24	PSet2, P3	L6, 5-7	L7, 21-38
How to solve	L4, 25-30	X	X	L8, 9-14
Solutions	L4, 31 L5, 11-13	PSet2, P3	L6, 8-11 L6, 17	L8, 15 L8, 21-25
QM Practice	L5, 14-20 PSet1, P2	X	L6, 12-16 PSet2, P2	L8, 16-20 PSet3, P2
Applications	L5, 21	X	L6, 18-37 L7, 10-19	L8, 26-34

Some Keywords

(Problem Set 2, Problem 1)

Quantum numbers

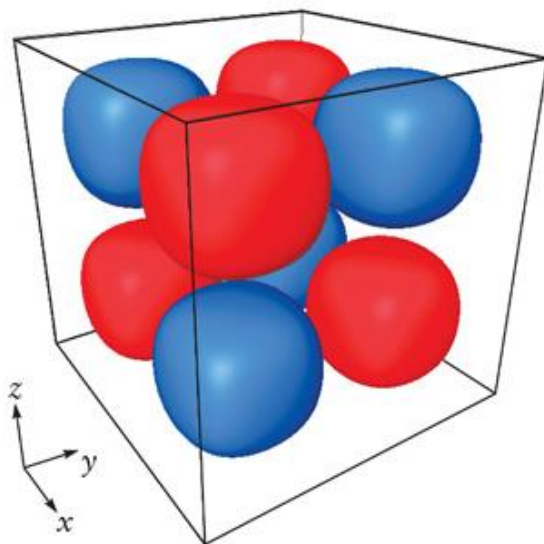
Degeneracy

Angular momenta



Visualization of Wavefunctions

3D PIB



RR

$(l, m) = (0, 0)$

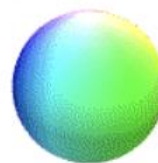


$$e^0 = e^{2i\pi}$$



$$e^{i\pi}$$

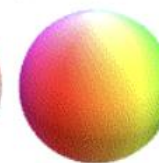
$(l, m) = (1, -1)$



$(l, m) = (1, 0)$



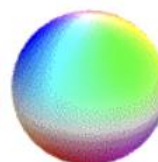
$(l, m) = (1, 1)$



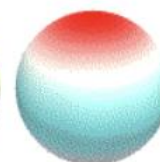
$(l, m) = (2, -2)$



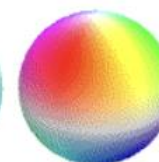
$(l, m) = (2, -1)$



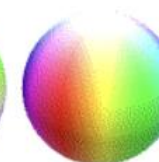
$(l, m) = (2, 0)$



$(l, m) = (2, 1)$



$(l, m) = (2, 2)$



(Images: Oxtoby et al., *Principles of Modern Chemistry*, 7th Ed. (2011), Fig. 4.33

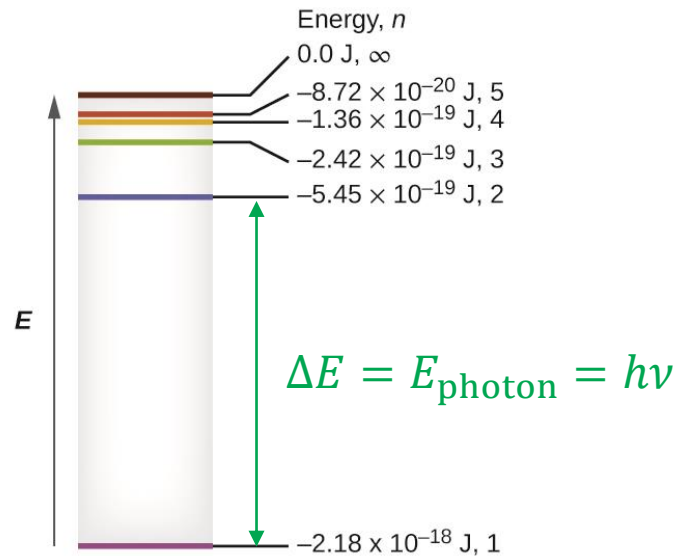
https://commons.wikimedia.org/wiki/File:Complex_Spherical_Harmonics_Figure_Table_Complex_Polar_Plot.gif

Basis of Spectroscopy

- Quantization of light

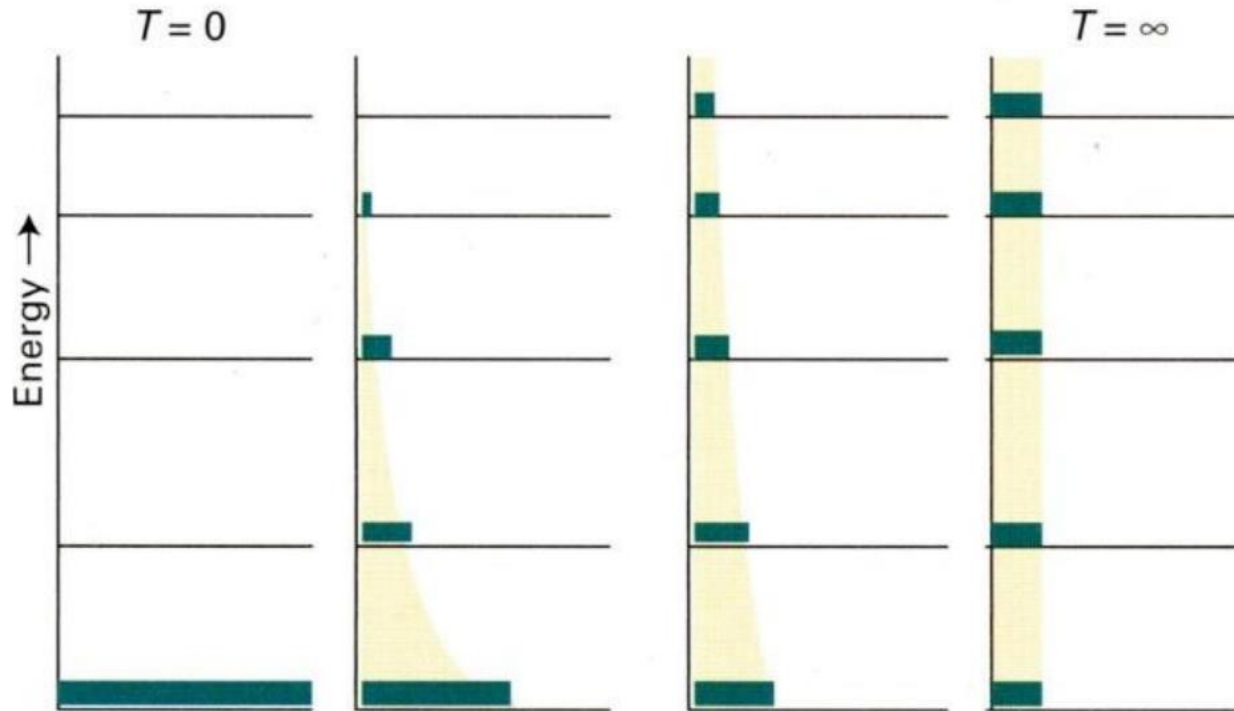
$$E = h\nu = hc/\lambda$$

- Quantization of system energy



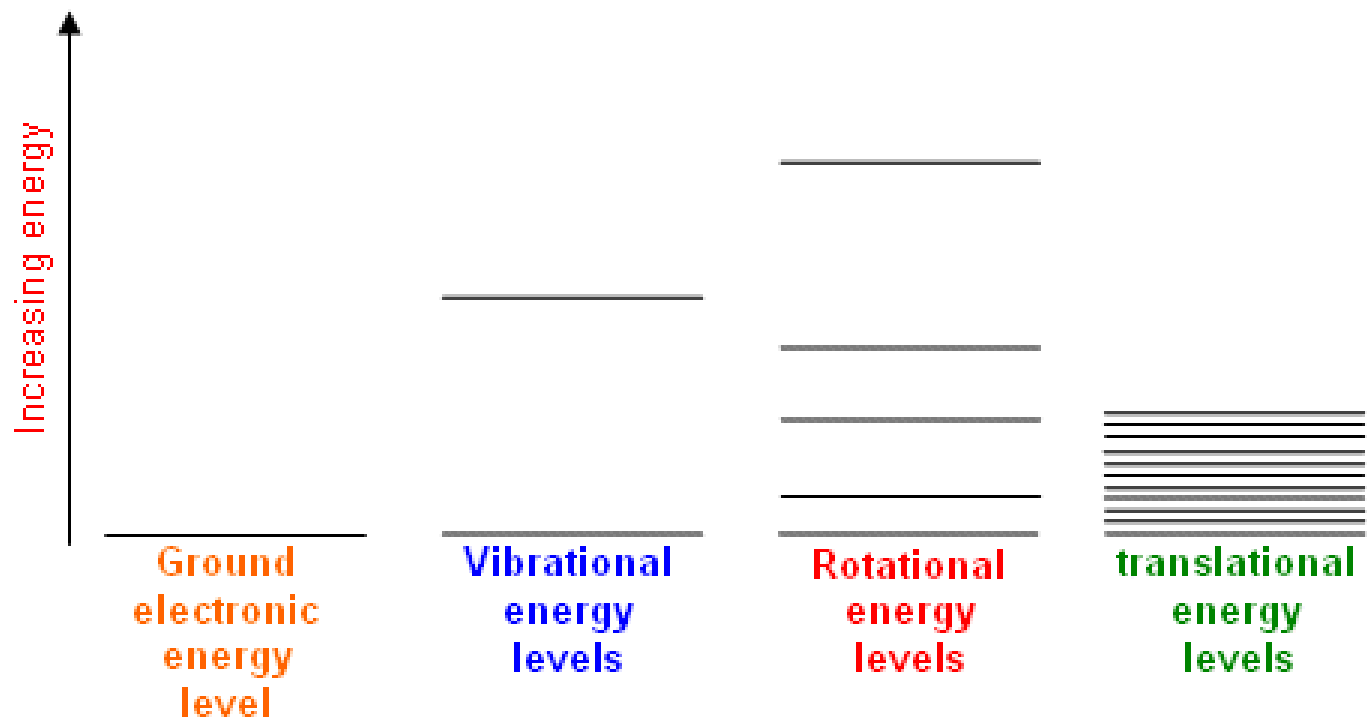
(Image: <https://opentextbc.ca/chemistry/chapter/6-2-the-bohr-model/>)

Thermal Excitation



$$P(E) \propto e^{-E/k_B T} \text{ (Boltzmann factor)}$$

Typical Energy Spacing



Selection Rules (PSet 2, P1)

Rules for the possible transitions of a system
from one quantum state to another

Simple harmonic oscillator

$$\Delta v = \pm 1$$

$$\frac{d\mu}{dx} \neq 0$$

Rigid rotor

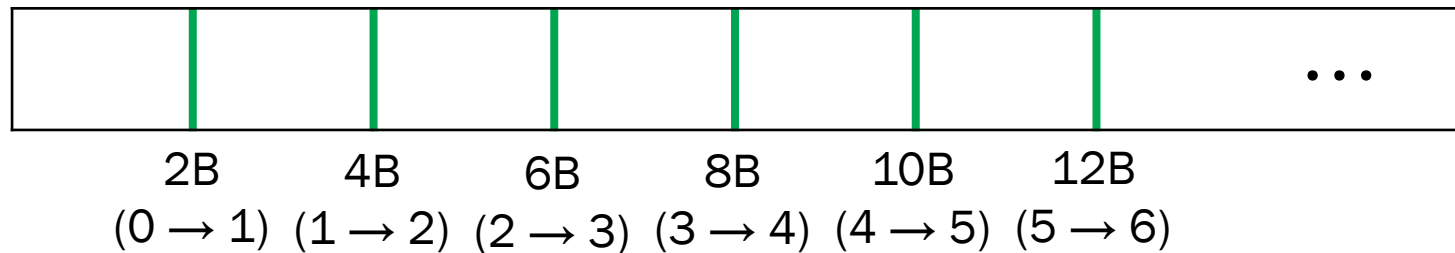
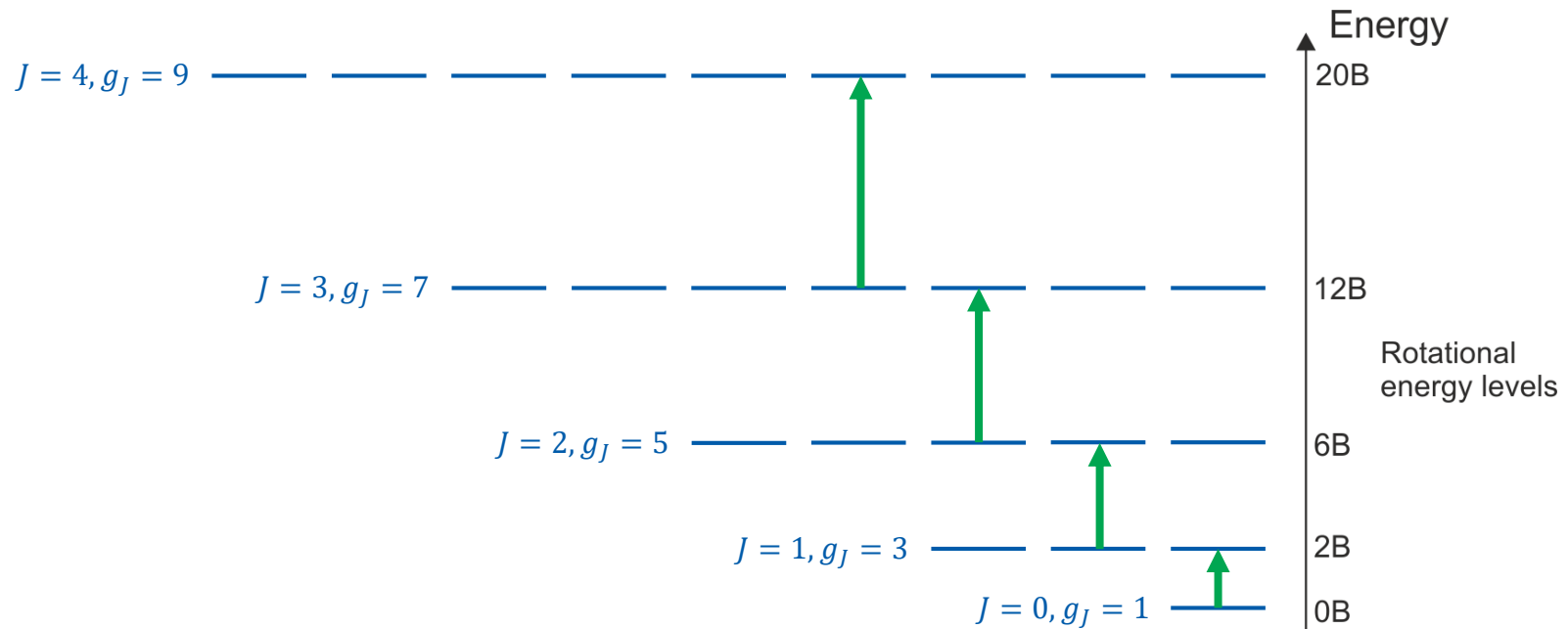
$$\Delta J = \pm 1$$

$$\mu \neq 0$$

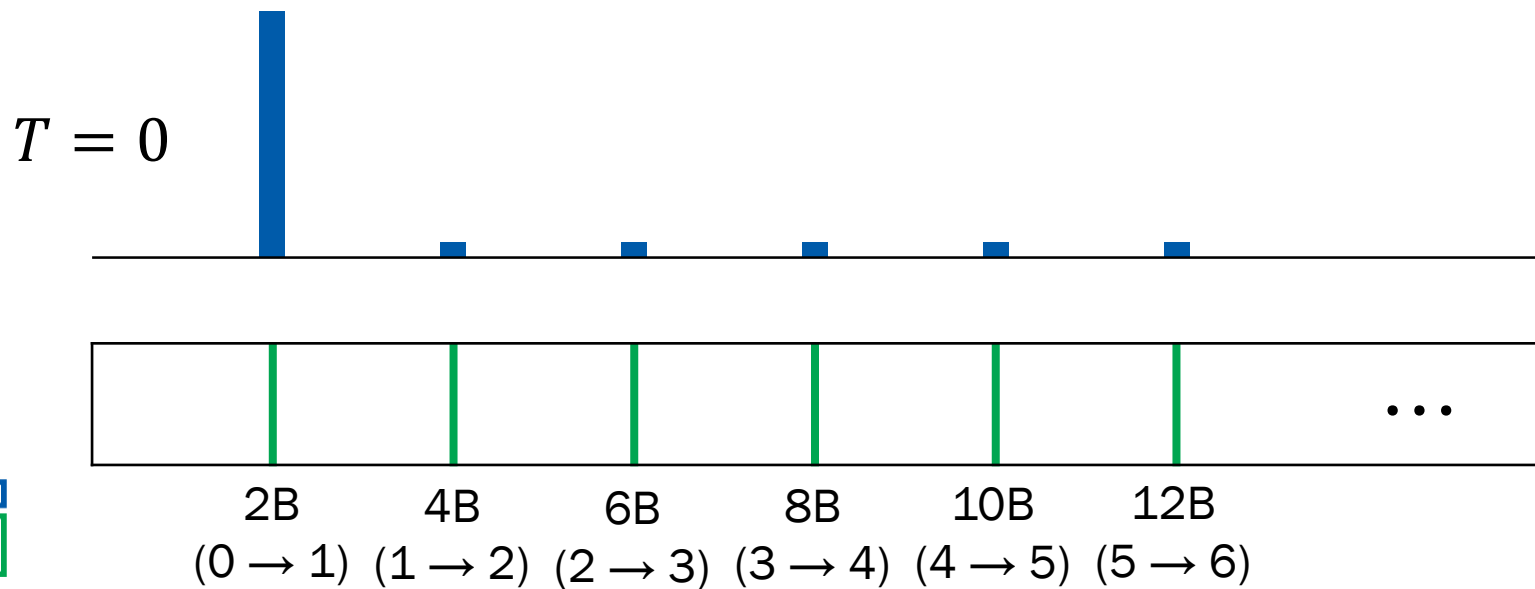
Applications

- **Molecular vibration:** infrared spectroscopy
 - Simple harmonic oscillator as a model
 - Validity of the model
 - Extension to polyatomic molecules
- **Molecular rotation:** microwave spectroscopy
 - Rigid rotor as a model
 - Validity of the model

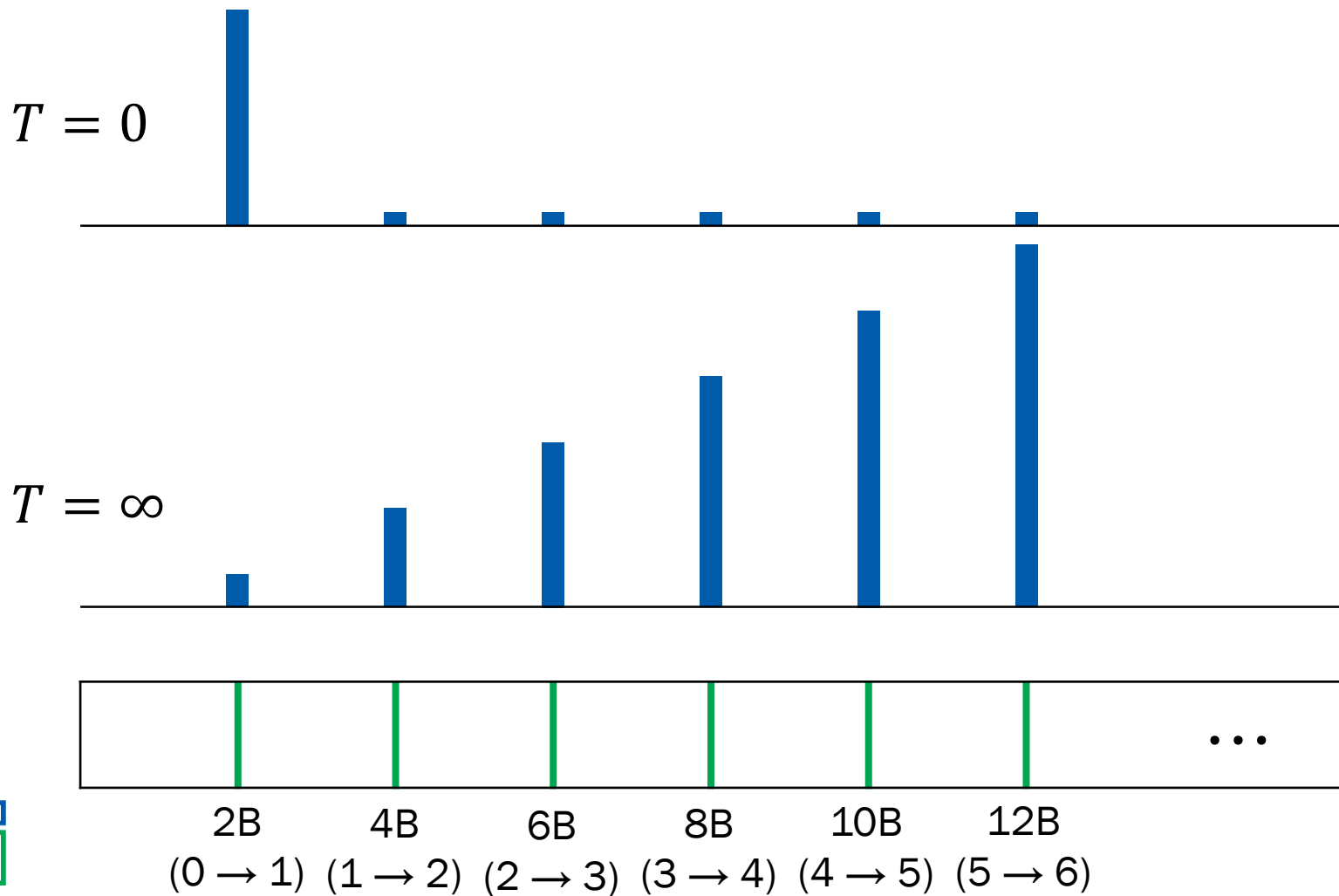
Rotational Spectrum



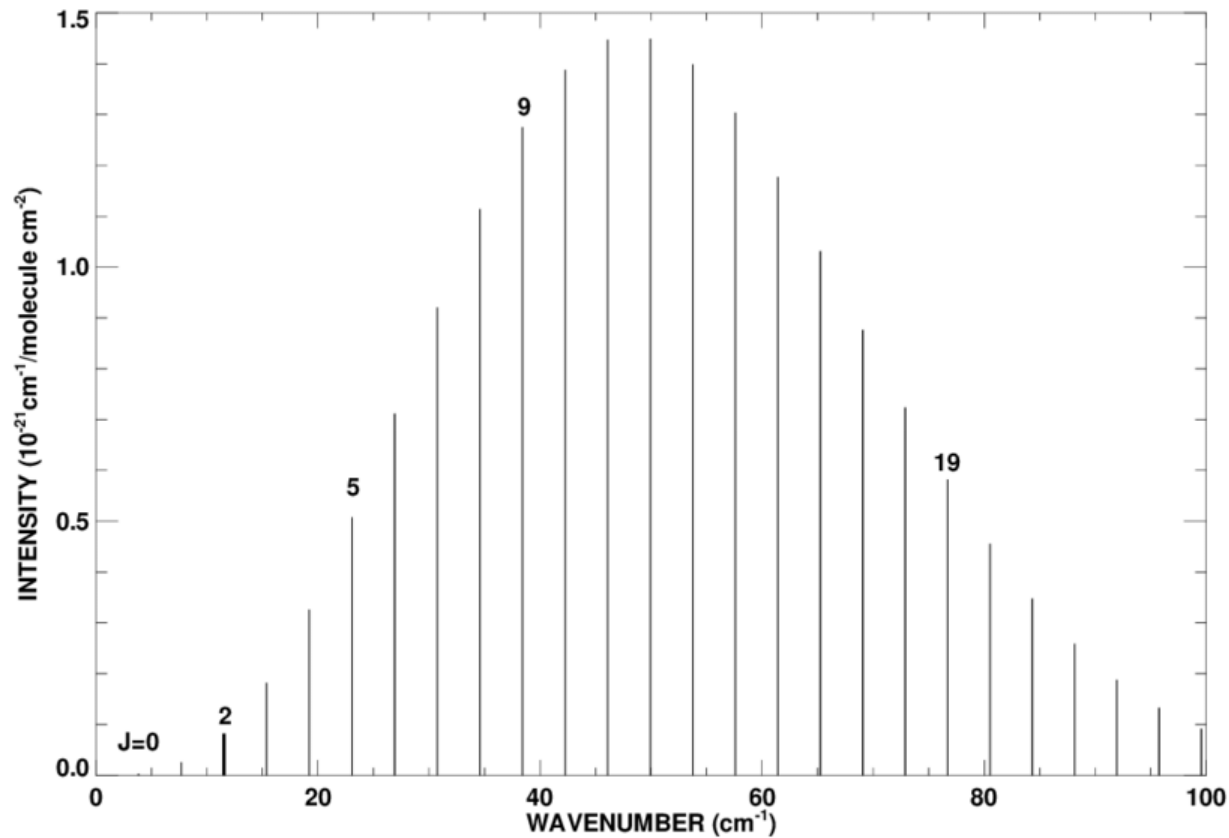
Populations?



Populations?



Microwave Spectrum of CO



(Image: https://www.researchgate.net/figure/The-rotational-spectrum-of-CO-The-line-intensities-are-for-a-temperature-of-296-K-The_fig10_253391065)

시험 범위 외

Nobel Prize in Physics 2023



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Pierre Agostini

Prize share: 1/3



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Ferenc Krausz

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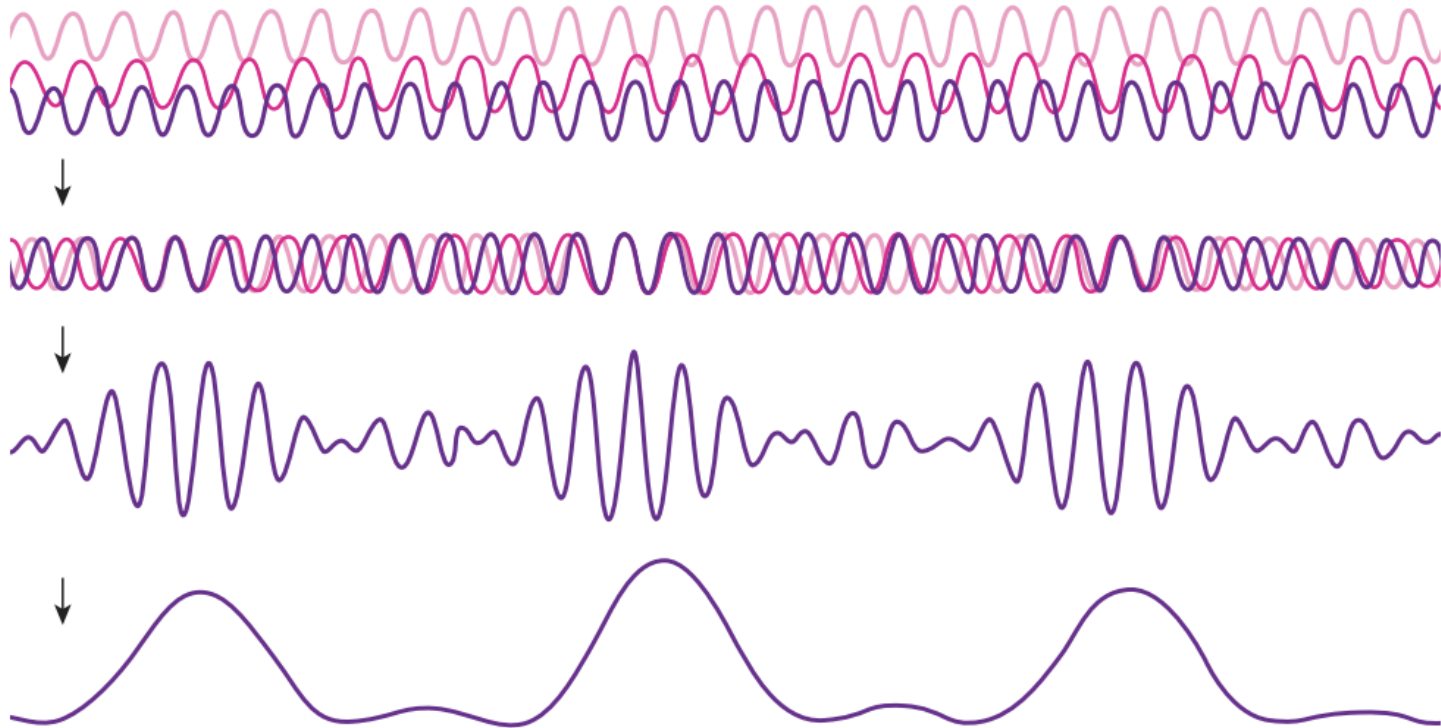
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Anne L'Huillier

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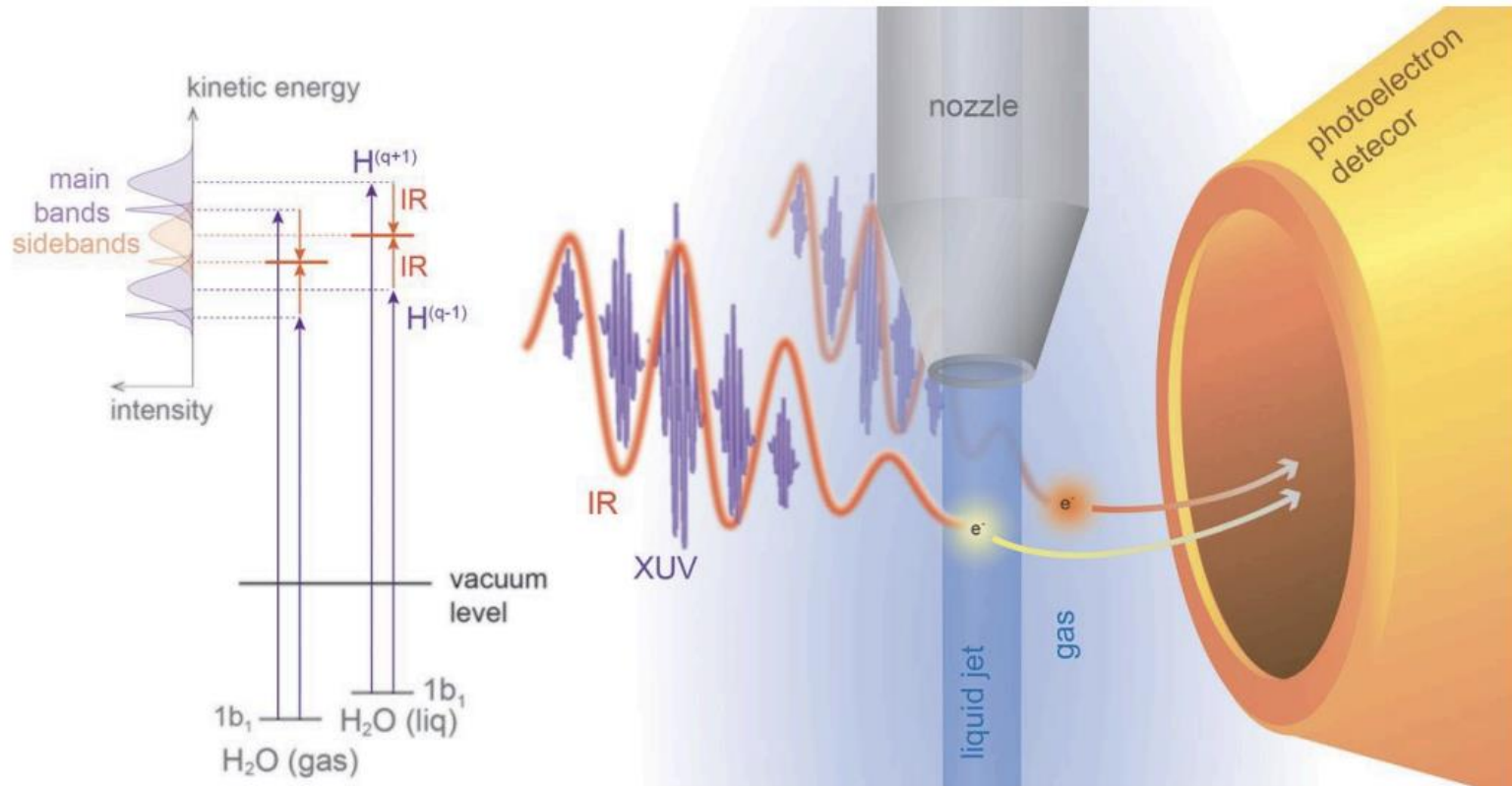
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